Absolute Error and MAE

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Absolute Error and MAE before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Mean, Median, and Baselines

Without opening the old lesson, explain one key idea from Mean, Median, and Baselines. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Error Analysis

Without opening the old lesson, explain one key idea from Error Analysis. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Confusion Matrix

Without opening the old lesson, explain one key idea from Confusion Matrix. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Absolute Error and MAE trace and record the first step you can explain without help.

Use W2: Compute squared **loss** for Absolute and squared **loss**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Absolute Error and MAE explanation against the supplied facts before discussing it.

Use the worked case for Absolute and squared **loss**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

Connect the Absolute Error and MAE target to a trace and a justified conclusion.

Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

WORKBOOK · 6 OF 21

Solve the given case

Solve the supplied Absolute Error and MAE case and show the deciding evidence.

Actual=[3,5,9], predicted=[2,7,6] cm.

Calculate MAE.

Show your trace, calculation or **evidence**

> **Write or draw your reasoning and evidence.**

WORKBOOK · 7 OF 21

Check a changed case

> Apply the procedure to the changed case and record why the conclusion follows.

Actual heights are 4, 8 and 10 cm; **prediction**s are 5, 6 and 13 cm. Calculate **absolute error**s and MAE.

My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 21 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

WORKBOOK · 9 OF 21

Project checkpoint

> Justify what Absolute Error and MAE adds to Responsible Plant Classifier and keep the evidence.

Use MAE correctly in the optional growth model; keep classifier and regression metrics clearly separated.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

Actual=[3,8,10], predicted=[5,5,11]. Compute signed residuals and MAE. A student averages residuals and gets zero; is that MAE?

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Compute MAE

Compute MAE. Apply the supplied Absolute Error and MAE facts and keep your reasoning.

Where to find the supplied example

Reference for this exercise: open this week's student lesson, "Absolute Error and MAE", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

Calculate MAE for the worked residuals.

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Compute squared loss

> Compute squared loss. Apply the supplied Absolute Error and MAE facts and keep your reasoning.

Give MSE and **RMSE** with **unit**s.

My reasoning and evidence

WORKBOOK · 14 OF 21

W3 Compare patterns

> Compare patterns. Apply the supplied Absolute Error and MAE facts and keep your reasoning.

Why do [0, 0, 6] and [2, 2, 2] have equal MAE but unequal MSE?

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply independently

Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Absolute Error and MAE facts and keep your reasoning.

Residuals [-2, 0, 4] m: calculate MAE MSE **RMSE** and the result of converting to cm.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Choose honestly

> Choose honestly. Apply the supplied Absolute Error and MAE facts and keep your reasoning.

A team switches from MSE to MAE only after seeing a bad large error. What should be reported?

My reasoning and evidence

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Absolute Error and MAE

> Compare absolute and squared error without losing track of units. Explain the labelled step before checking the changed condition.

Compare absolute and squared error without losing track of **unit**s.

Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

AI ACADEMY / AI-4/21

Absolute Error and MAE

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Absolute errors

Absolute values1, 2, 3 give MAE6/3=2 cm.

2. Squared errors

Squares1, 4, 9 give MSE14/3 cm2, approximately = 4.67 cm2.

3. Root squared error

RMSE=sqrt(14/3), approximately2.16 cm.

CHANGE ONE THING

Convert every residual to millimetres.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Evaluate and revise. Compare with your starting method, explain errors and record a justified revision.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Compare absolute and squared error without losing track of **unit**s.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? **RMSE**=sqrt(14/3), approximately2.16 cm.

1. **Absolute error**s

2. Root squared error

3. Squared errors

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Root squared error **RMSE**=sqrt(14/3), approximately2.16 cm. Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm.

The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Convert every residual to millimetres. Which conclusion is supported by the supplied facts?

1. MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

2. This one example guarantees correct results for every future **input**.

3. We can decide the result without examining the changed **input** or the procedure.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm. Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Repair a missing building block

Review Mean, Median, and Baselines (ai-4/20). Goal: Fit mean and median constant baselines on **training** targets. Short practice: Does either original baseline win under evaluation MAE?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Error Analysis (ai-4/18). Goal: Categorize development errors using **evidence**. Short practice: What **evidence** would support the claim that a change helps fresh dim-light images?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Confusion Matrix (ai-4/15). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Why is adding TP alone without reducing FN an incorrect revision?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

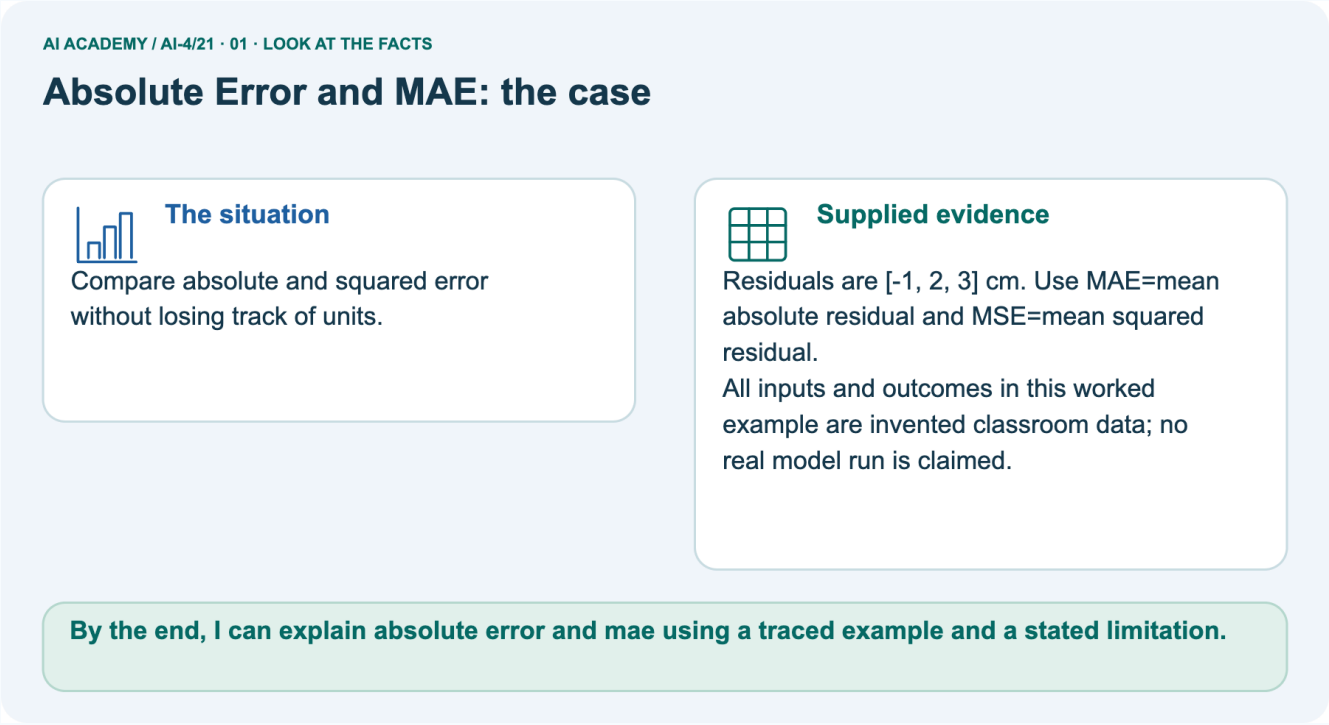
Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

Visual concept journey

> Explain each Absolute Error and MAE picture from its labels, then record a relevant limit.

01. Absolute Error and MAE: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

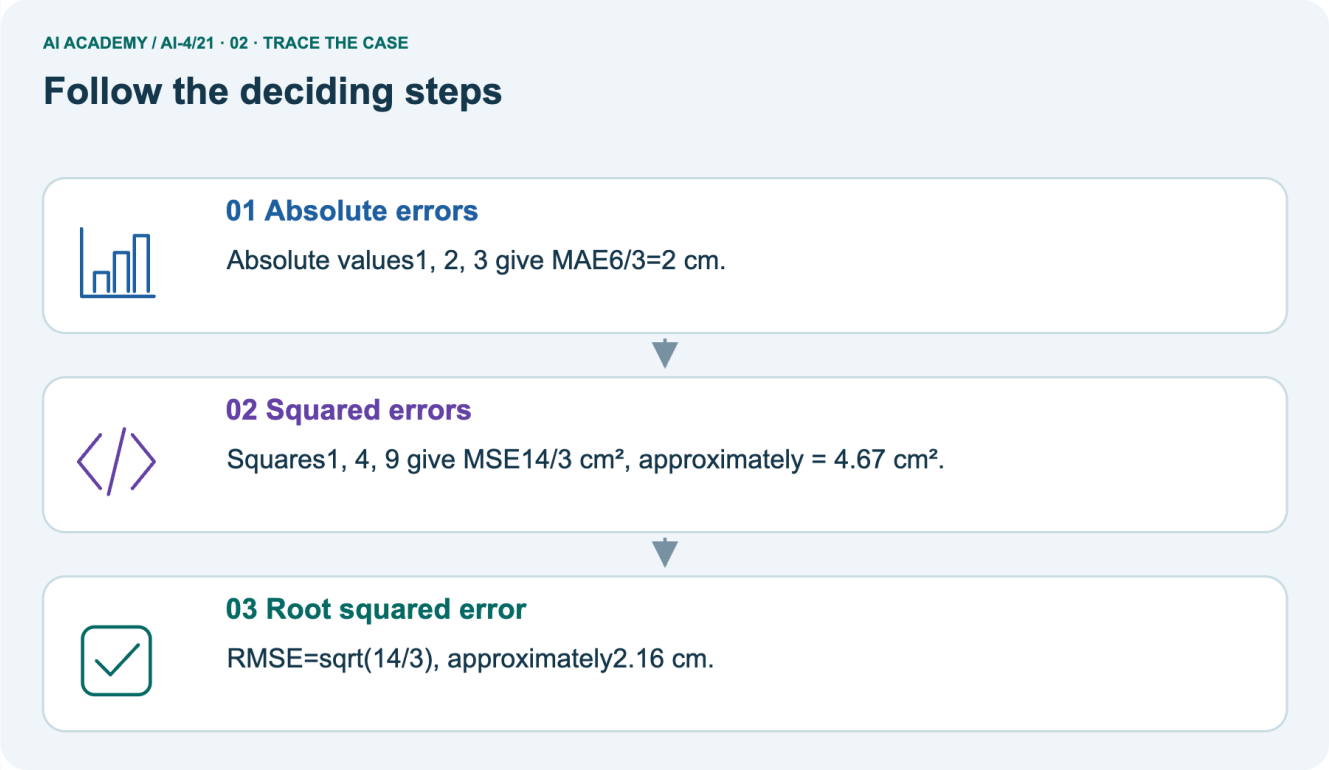
Compare absolute and squared error without losing track of **unit**s. Supplied facts: Residuals are [-1, 2, 3] cm. Use MAE=mean absolute residual and MSE=mean squared residual.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



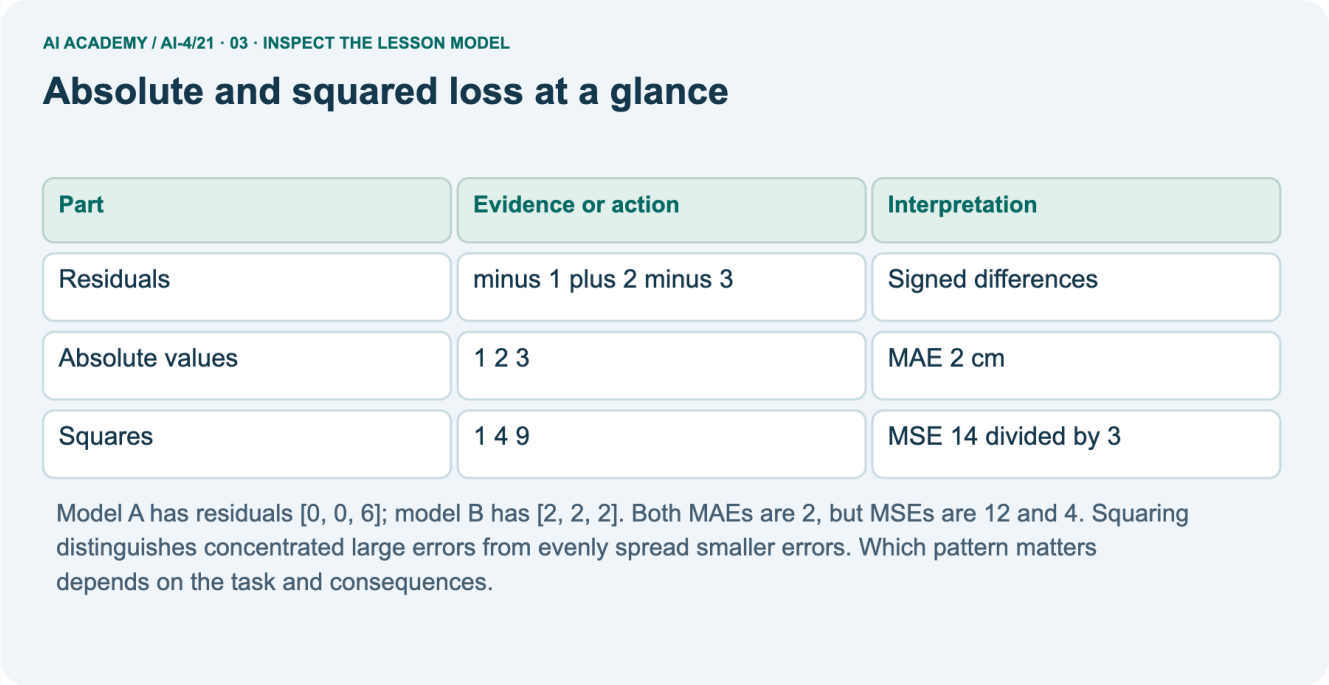
Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute errors: Absolute values1, 2, 3 give MAE6/3=2 cm. → Squared errors: Squares1, 4, 9 give MSE14/3 cm², approximately = 4.67 cm². → Root squared error: **RMSE**=sqrt(14/3), approximately2.16 cm.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Absolute and squared loss at a glance



Follow the labelled picture with the matching explanation. Open the vector diagram

Absolute and squared **loss** at a glance. Residuals | minus 1 plus 2 minus 3 | Signed differences. Absolute values | 1 2 3 | MAE 2 cm. Squares | 1 4 9 | MSE 14 divided by 3

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / AI-4/21 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

RMSE=sqrt(14/3), approximately2.16 cm.

Changed condition

Convert every residual to millimetres.

Worked changed-case result

MAE becomes20 mm; MSE becomes1400/3 mm²; RMSE becomes about21.60 mm.
Why: Residuals scale by10; squares scale by100, while the square root returns a length unit.

Follow the labelled picture with the matching explanation. Open the vector diagram

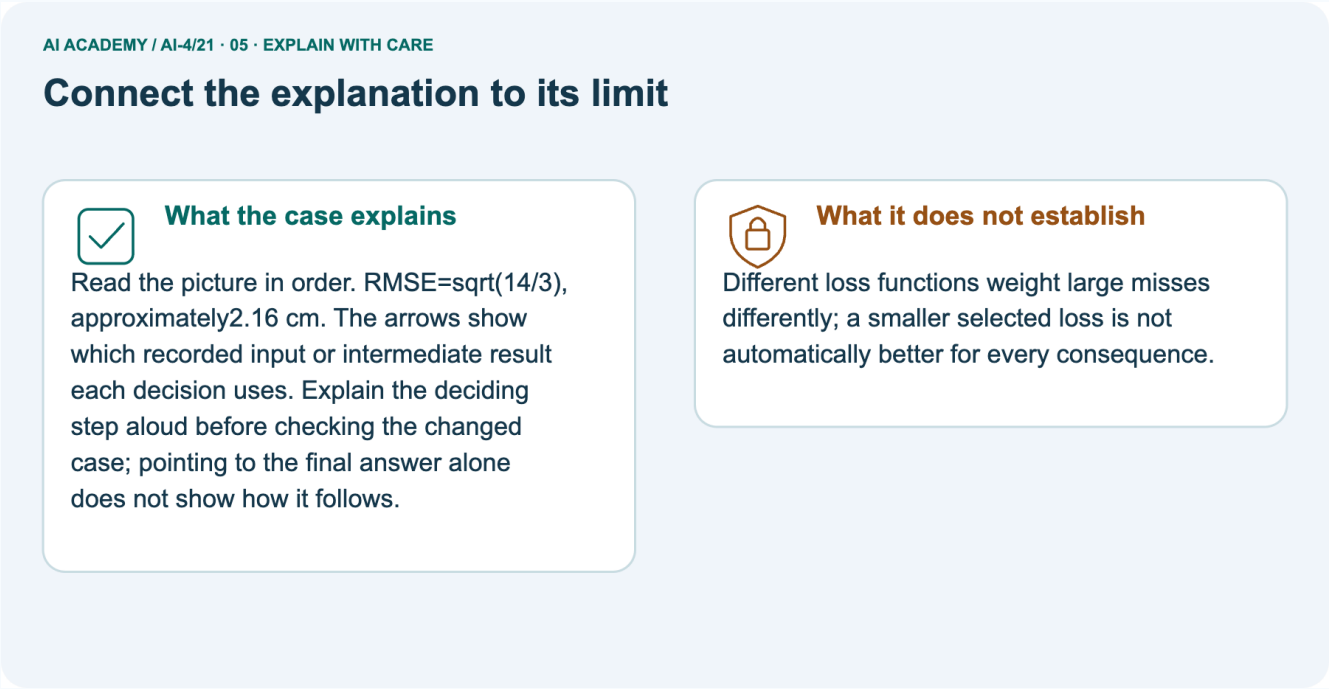
Original case: **RMSE**=sqrt(14/3), approximately2.16 cm. Changed condition: Convert every residual to millimetres. Expected result: MAE becomes20 mm; MSE becomes1400/3 mm²; **RMSE** becomes about21.60 mm.

Reason: Residuals scale by10; squares scale by100, while the square root returns a length **unit**.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. **RMSE**=sqrt(14/3), approximately2.16 cm. The arrows show which recorded **input** or intermediate result each decision uses.

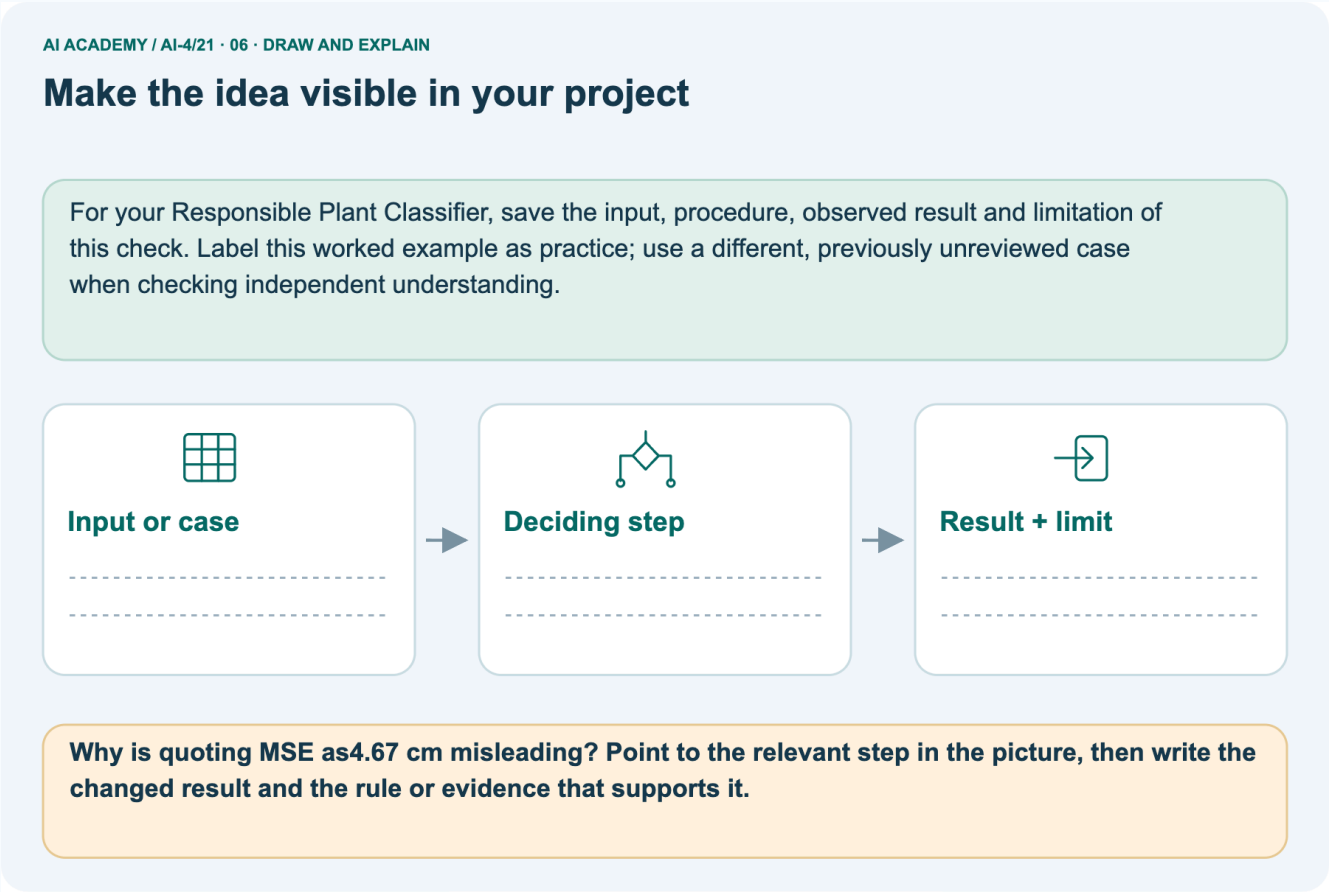
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Different **loss** functions weight large misses differently; a smaller selected **loss** is not automatically better for every consequence.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why is quoting MSE as4.67 cm misleading? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.